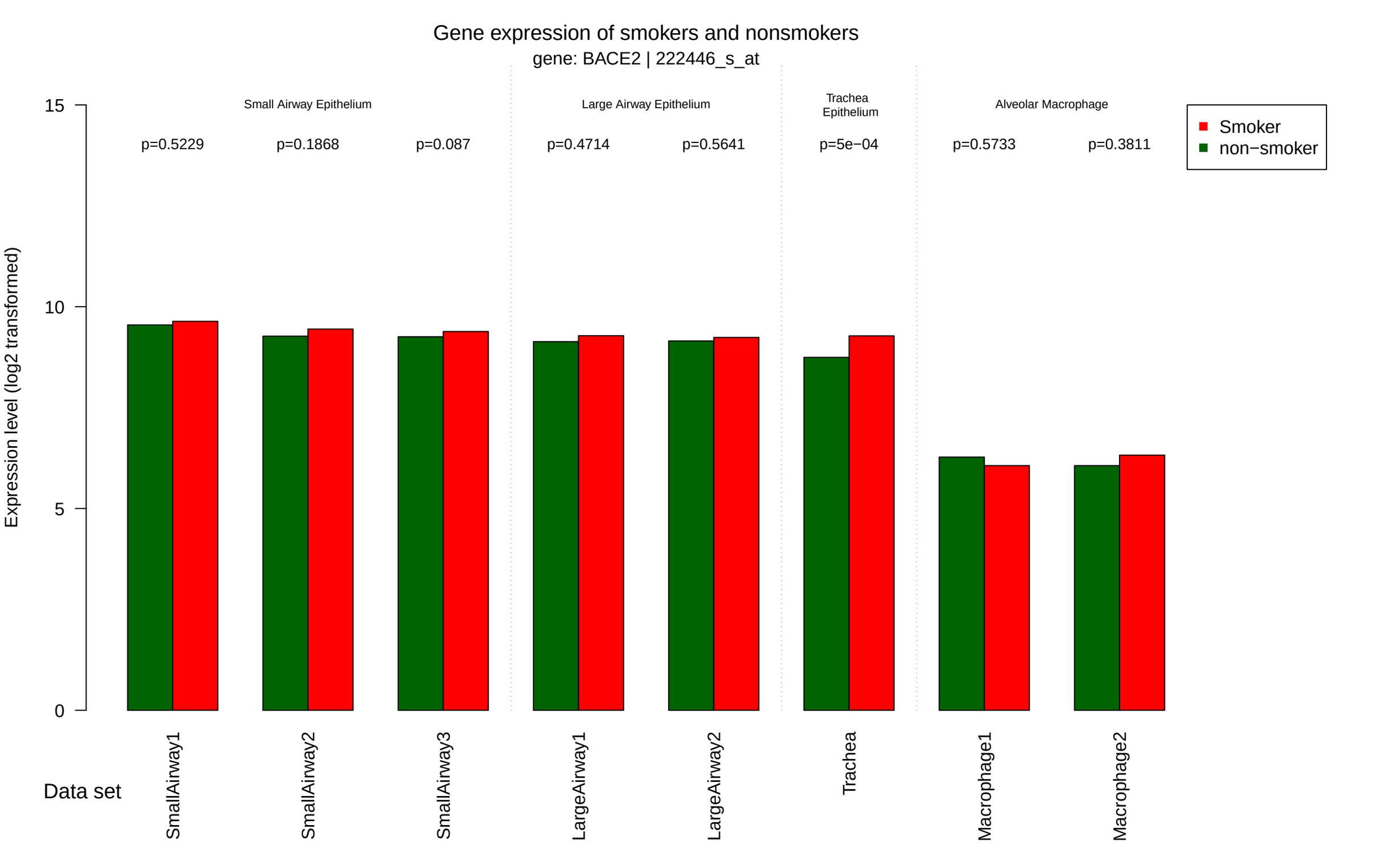
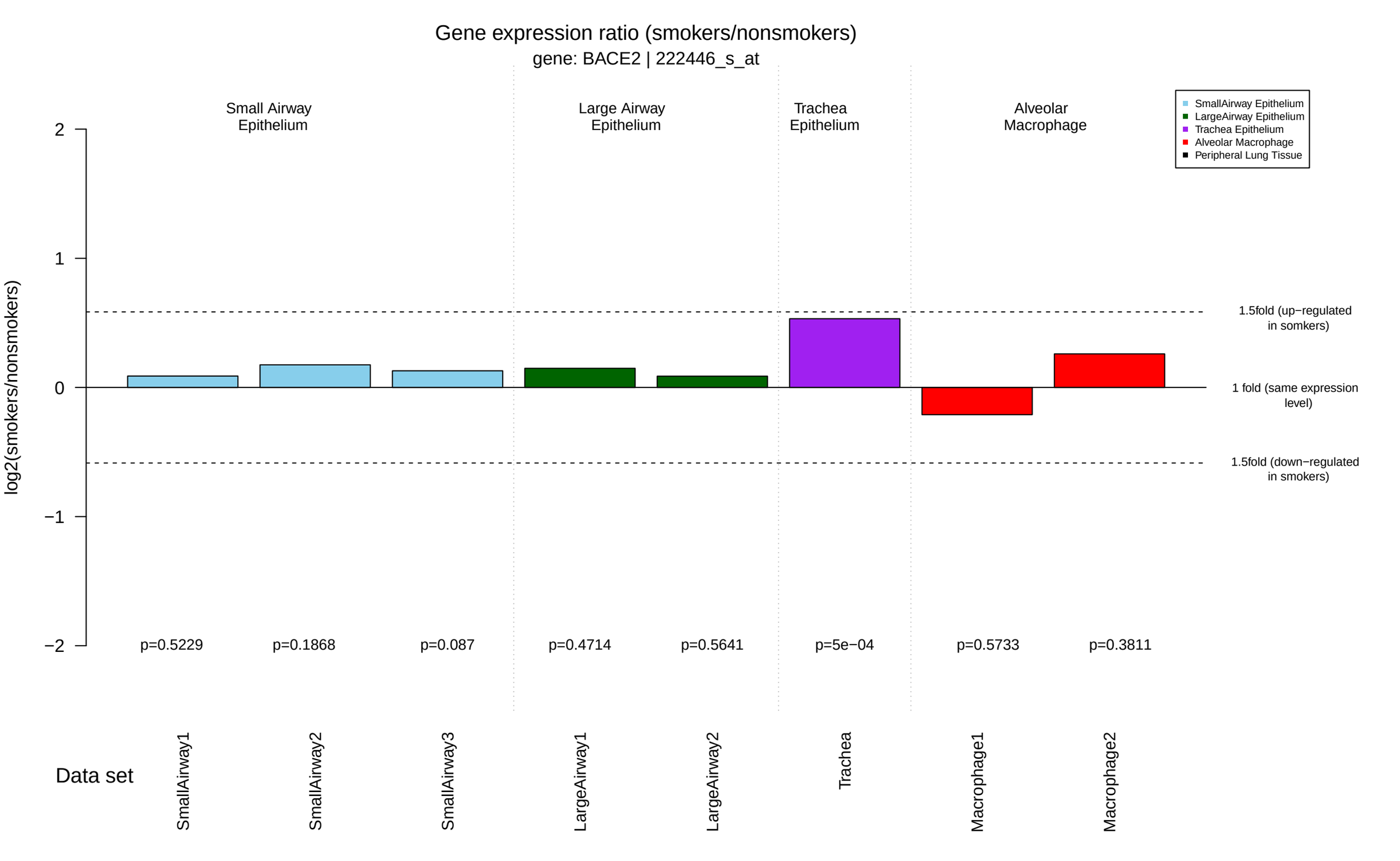
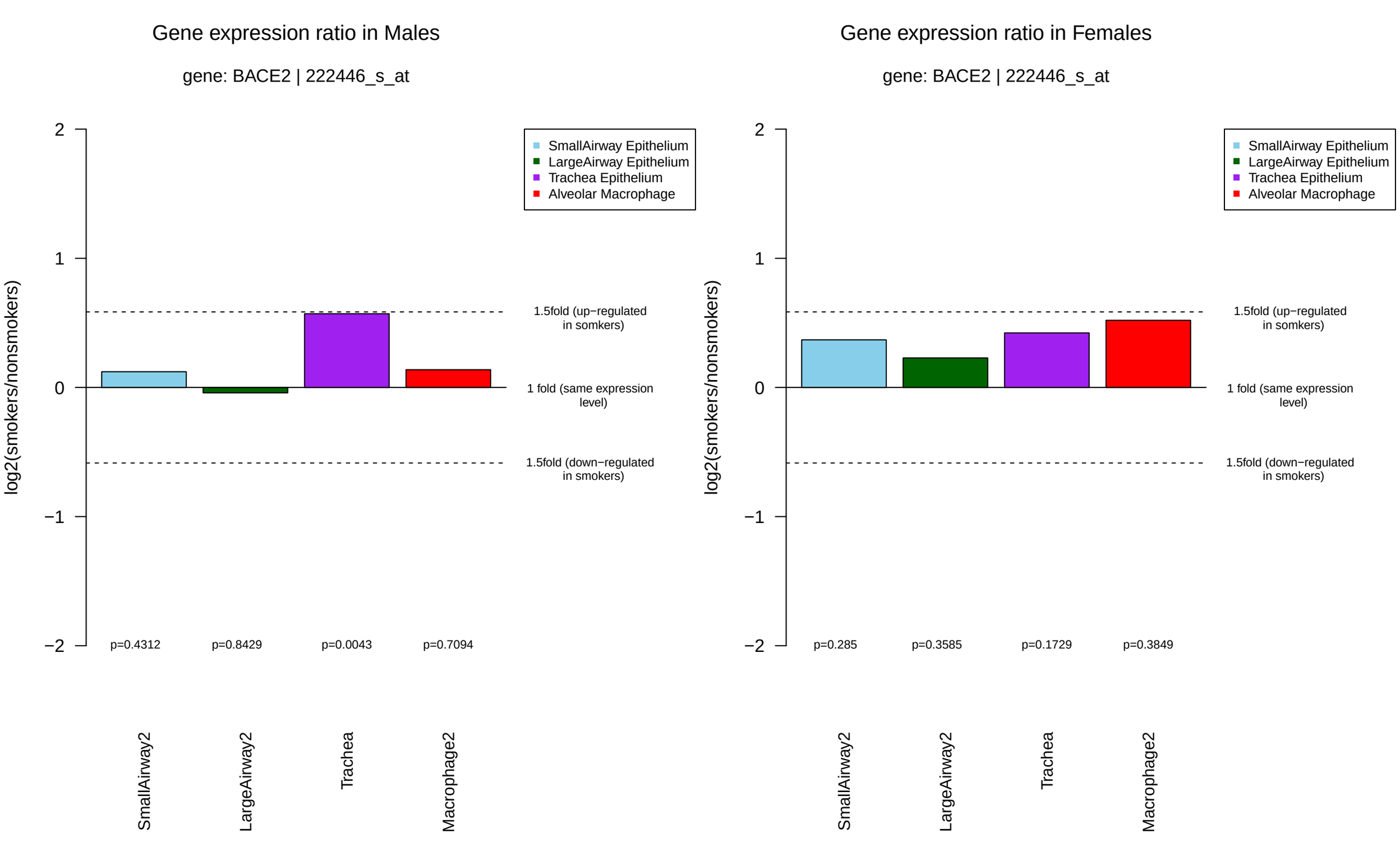
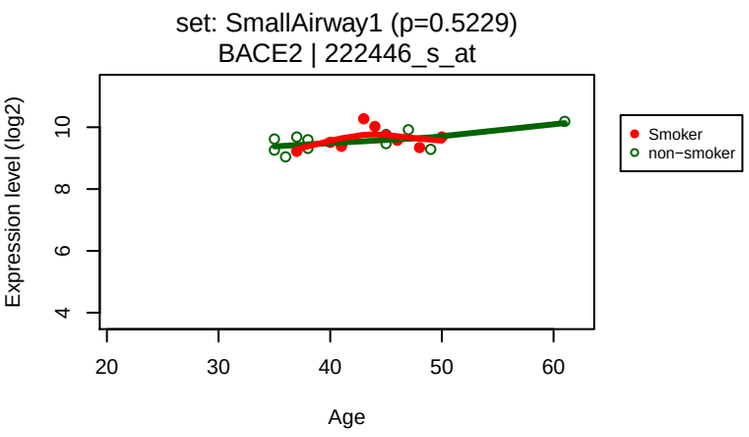
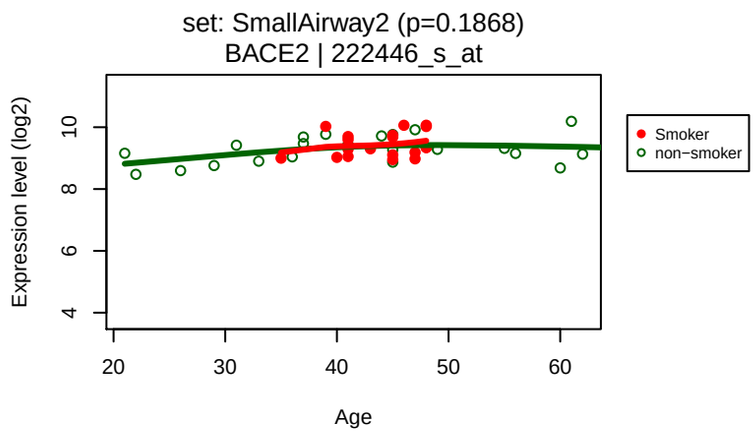
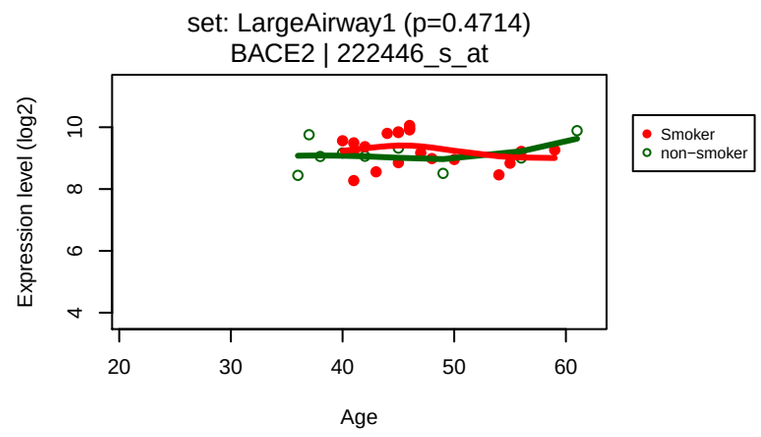

Gene expression of smokers and nonsmokers
gene: BACE2 | 222446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5229
p=0.1868
p=0.087
p=0.4714
p=0.5641
p=5e−04
p=0.5733
p=0.3811
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

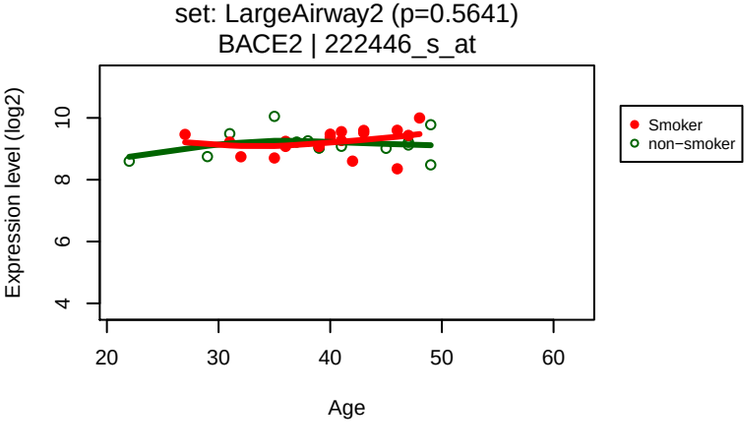

Gene expression ratio (smokers/nonsmokers)
gene: BACE2 | 222446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5229
p=0.1868
p=0.087
p=0.4714
p=0.5641
p=5e−04
p=0.5733
p=0.3811
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

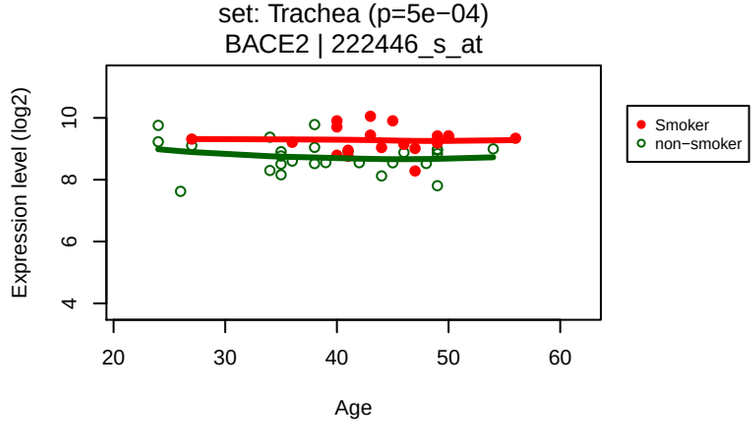

Gene expression ratio in Males
gene: BACE2 | 222446_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4312
p=0.8429
p=0.0043
p=0.7094
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACE2 | 222446_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.285
p=0.3585
p=0.1729
p=0.3849
SmallAirway2
LargeAirway2
Trachea
Macrophage2

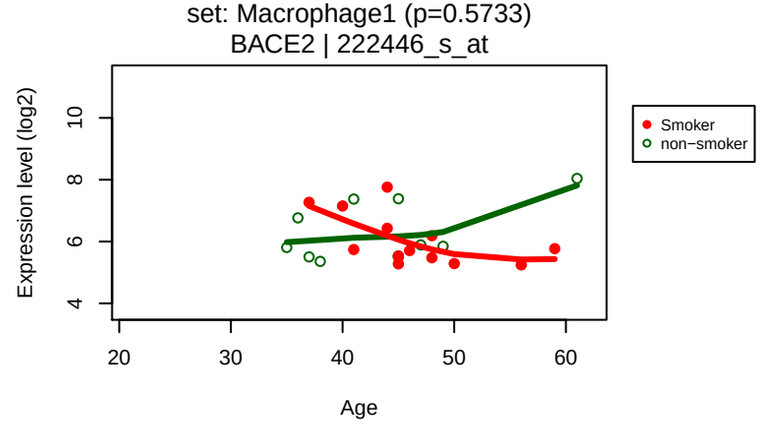

set: SmallAirway1 (p=0.5229)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker

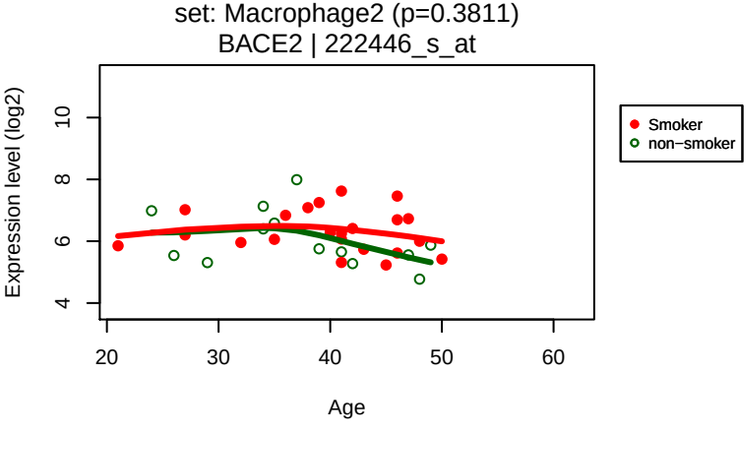

set: SmallAirway2 (p=0.1868)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker

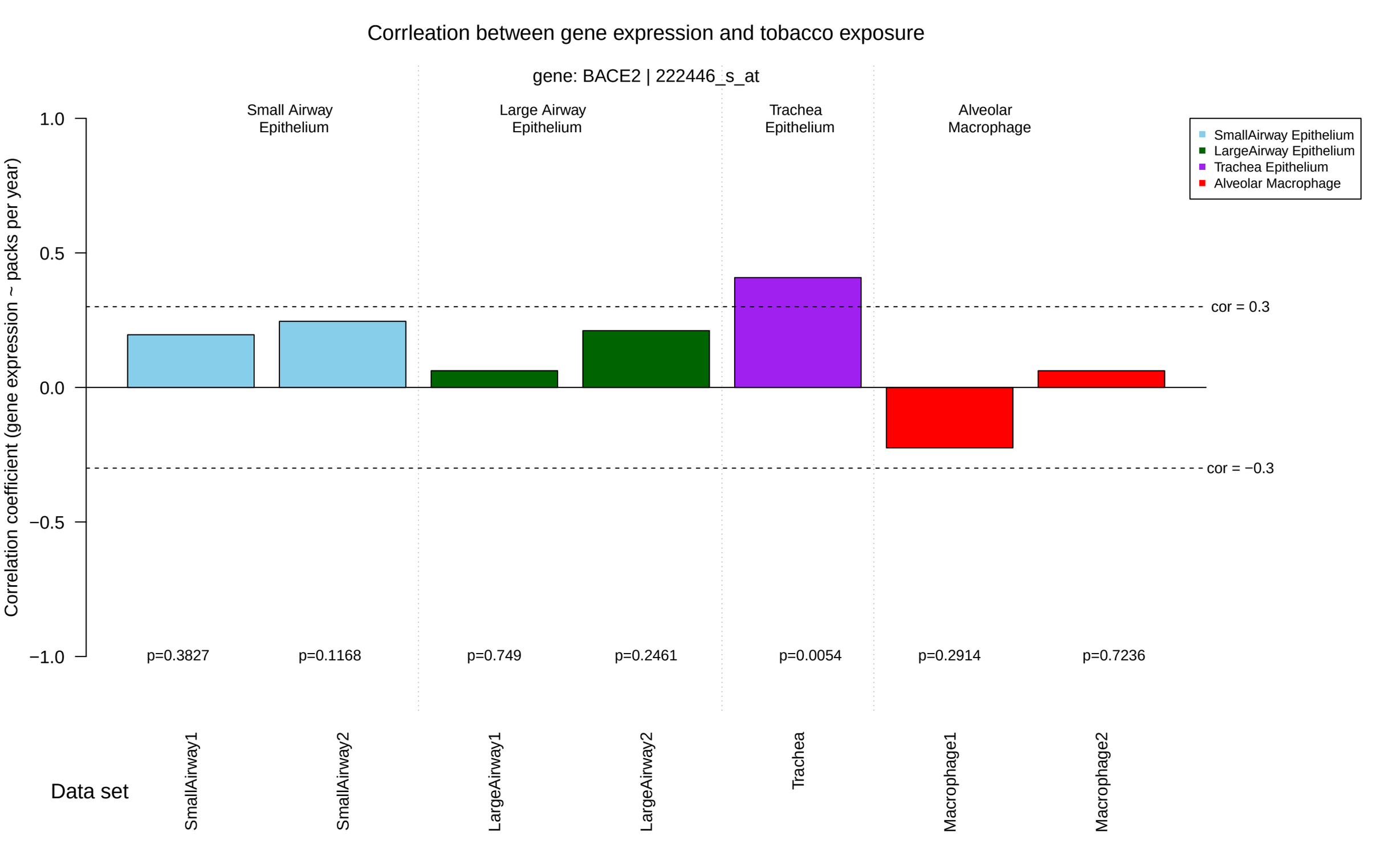

set: LargeAirway1 (p=0.4714)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5641)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=5e−04)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5733)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3811)
BACE2 | 222446_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACE2 | 222446_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3827
p=0.1168
p=0.749
p=0.2461
p=0.0054
p=0.2914
p=0.7236
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2